



BEST PRACTICE SERIES

Pressure OHT Air Induction System to Check for Leaks

Maintenance
and Repair

Application

Component
Life
Management

Component
Renewal
(CRC)

MARC
Management

Pressure OHT Air Induction System to Check for
Leaks0

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1.0 Introduction

Condition Monitoring (CM) is a critical component to any successful Maintenance & Repair (M&R) repair-before-failure strategy. The foundation of any successful Condition Monitoring program is regular and thorough machine inspections. Discipline to conduct inspections and the attention to detail to detect potential problems or existing defects is very important. Defect detection must be followed by prompt and appropriate execution by the Backlog Management and Planning & Scheduling departments.

Dust entry through the air induction system and into the engine can cause significant downtime, productivity loss, customer frustration, and significant man-hours to repair or replace the engine.

2.0 Best Practice Description

Pressurizing the air induction system can allow the maintenance team to check for proper integrity and sealing of the induction tubes, hoses and clamps. Early detection of problems or potential problems is very important. Use machines with known healthy air systems to establish site specific baselines for silicone and aluminum in engine oil SOS. This will aid SOS interpreters in identifying air systems issues.

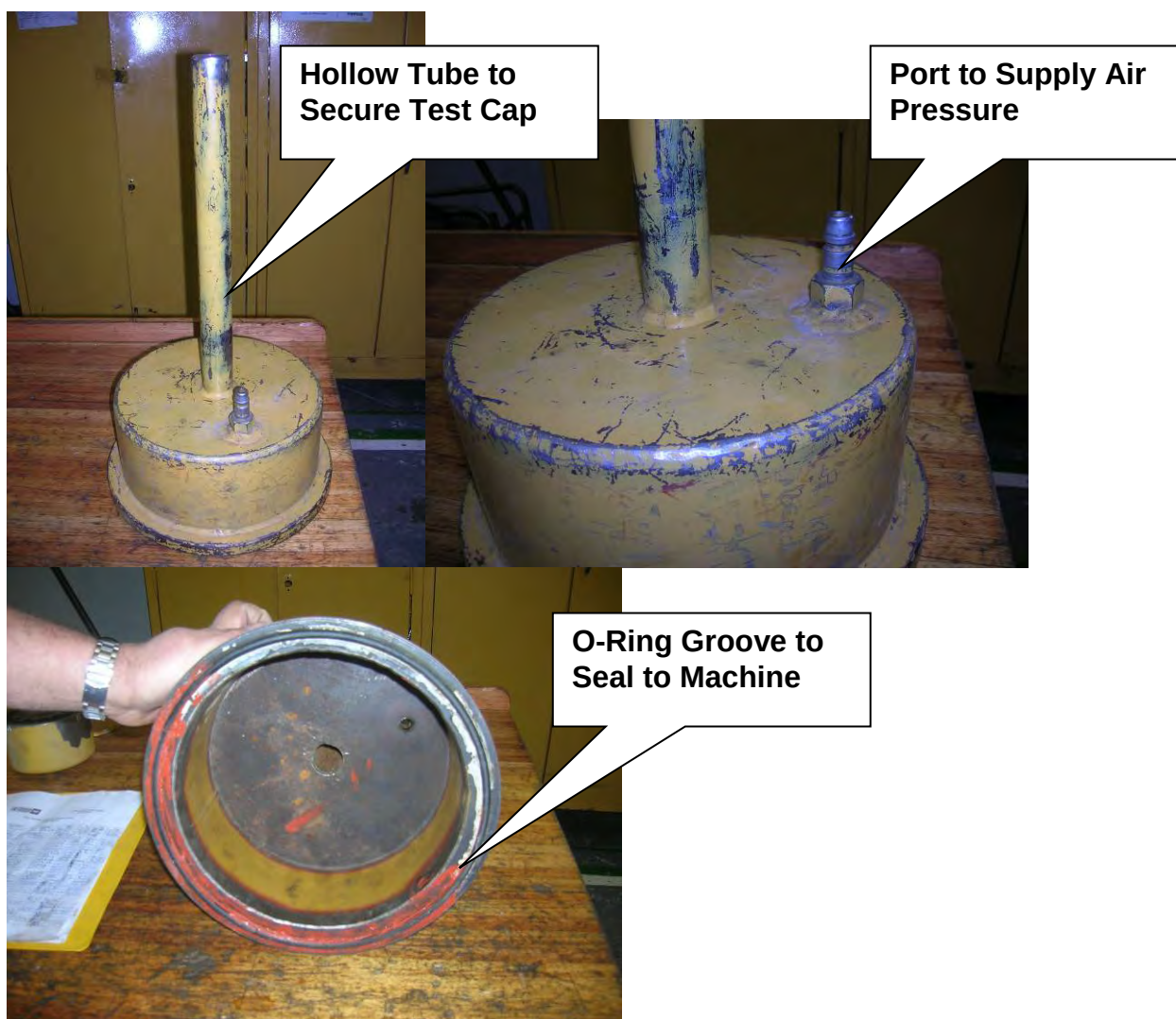
3.0 Implementation Steps

1. Identify service interval to employ this test. (For instance: Every 500 hour PM, etc).
2. Employ lockout and tag-out procedure.
3. Remove air filters.
4. Install test caps to all filter housings and secure in place using threaded nut.
5. Pressurize test cap using clean and dry shop air.
 - a. Suggest air pressure to 4 psi (28 kpa).
6. Listen for any sounds of air leakage from the system.
 - a. Good: No sound of air leakage.
 - b. Not Good: Sound of air leakage. If sound of air leakage, identify air source and take appropriate action to resolve.
7. Spray mild dilution of soapy water on joints and rubber hoses in air induction system.
8. Visually inspect for bubbles (bubbles indicate the presence of a leak in that location).
 - a. Good: No soapy bubbles (machine has a well sealed induction system).
 - b. Not Good: If air bubbles are seen. If so, identify and take appropriate action to resolve.

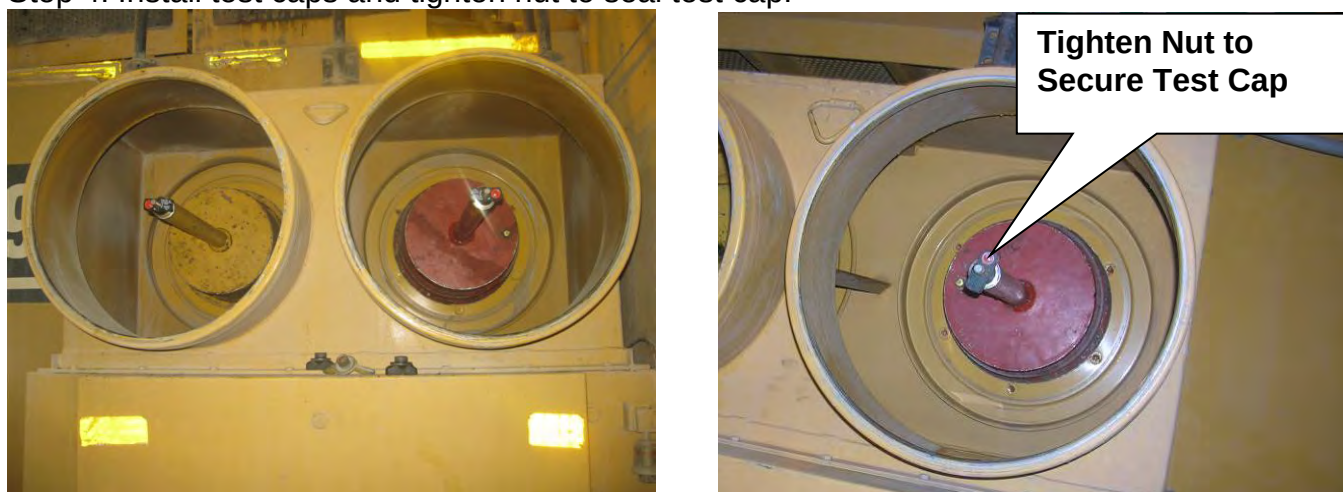
Photos of Test Caps to Test Air Induction System for Leaks. Design features are:

- 1) Long hollow tube to allow cap to be secured in place of air filter
- 2) Test port to supply air pressure to cap during pressure testing
- 3) O-ring groove to seal to machine and prevent air leakage

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**Basic directions to use the test caps to pressurize the system:**

Step 4: Install test caps and tighten nut to seal test cap.



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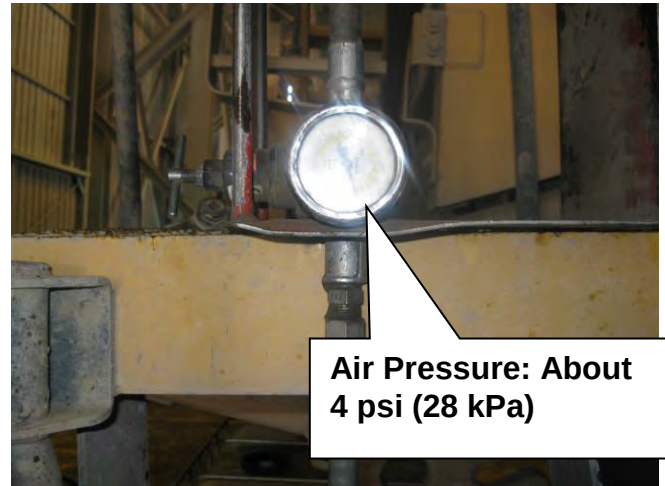
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Step 5: Pressurize air induction system.



Step 6: Spray mild soapy water on joints, hoses, and clamps.



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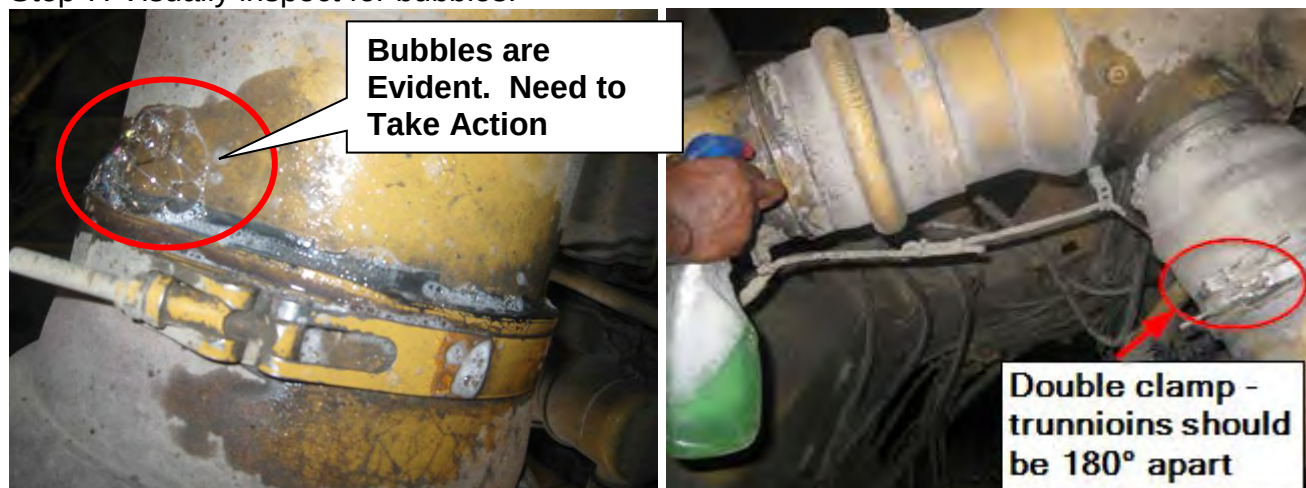
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Step 7: Visually inspect for bubbles.



4.0 Benefits

Detection of air induction pressure loss and / or loose induction clamps can have significant benefits. Early detection is key to limiting severity of failure, repair cost of parts and labor, and machine downtime to resolve.

5.0 Resources Required

The major resource required to implement this Best Practice is the fabrication of the test caps to install on the air filter boxes.

Fabrication costs are approximately US \$ 1000 / per set of four.

Parts List is as Follows:

- 1) O-ring Seal: 096-0037 (ID: 264.5 mm, Cross Section: 8.4 mm)
- 2) Wing Nut: 1N-8161
- 3) Material for Test Cap: Aluminum

6.0 Supporting Attachments / References

- 793 Drawing
- SOS Interpretation 1 help sheets pg 12

7.0 Related Best Practices

Best Practice **0107-2.15-1059 “Using VIMS® to Determine Air Filter Replacement Intervals”** explains how to use VIMS® to check your air filter performance.

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